

# JIAHAO XIA

College Park, MD | [jxia77@terpmail.umd.edu](mailto:jxia77@terpmail.umd.edu)  
<https://jjjxia.github.io/>

## EDUCATION

---

|                                                            |                     |
|------------------------------------------------------------|---------------------|
| <b>University of Maryland</b> , College Park, MD           | Aug 2025 – May 2028 |
| B.S. in Computer Science                                   | GPA: 3.73/4.00      |
| Minor in Mathematics                                       |                     |
| <b>Pennsylvania State University</b> , University Park, PA | Aug 2024 – May 2025 |
| Electrical Engineering coursework (transferred)            | GPA: 3.94/4.00      |

## RESEARCH INTERESTS

---

Automated reasoning, formal verification, program analysis

## PUBLICATIONS

---

- Xuyang Li\*, **Jiahao Xia\***, Ahmed Adnan, and Jingbo Wang. *pSoufflé: Towards Scalable Probabilistic Logic Inference for Program Analysis*. Submitted to Formal Methods in Computer-Aided Design, 2026.

## RESEARCH EXPERIENCE

---

|                           |                    |
|---------------------------|--------------------|
| <b>Purdue University</b>  | May 2025 – Present |
| <i>Research Assistant</i> | Remote             |

### Scalable Probabilistic Logic Inference for Program Analysis

- Extended the **Soufflé Datalog engine** in **C++** to support probabilistic facts, queries, and evidence.
- Integrated **Sentential Decision Diagrams** and **weighted model counting** for exact probabilistic inference.
- Built the evaluation pipeline, including **vProbLog/ProbLog** baseline integration and artifact.

## SELECTED PROGRAMS & FELLOWSHIPS

---

- |                                                |      |
|------------------------------------------------|------|
| • PLMW Scholarship, <i>PLDI'26</i>             | 2026 |
| • Fifteenth Summer School on Formal Techniques | 2026 |
| • Oregon Programming Languages Summer School   | 2026 |

## RELEVANT COURSEWORK

---

**Computer Science:** Computer Systems, Discrete Mathematics, Data Structures, Algorithms, Data Science, Programming Languages

**Mathematics:** Introductory and Advanced Linear Algebra, Multivariable Calculus, Complex Analysis, Probability and Statistics

## TECHNICAL SKILLS

---

- **Tooling:** Linux, Git, Soufflé, ProbLog, C/C++, Python, Java, Ocaml, Rocq Prover